

Lecture 8: Computability

Alphabets, Strings, and Languages

Alphabet: A nonempty *finite* set Σ of symbols (characters).

$\Sigma = \{0,1\}$ is the default choice.

String: A *finite* sequence of zero or more symbols.

(or “word”)

The *empty* string (having zero symbols) is denoted ϵ .

For any $a \in \Sigma$:

a^k means ____ a's

a^* means ____ a's

a^+ means ____ a's

Σ^k = set of _____ strings over Σ .

Σ^* = set of _____ strings over Σ .

Σ^+ = set of _____ strings over Σ .

For any x, y : $x|y$ means _____

Language: A collection (set) of strings.

I.e., any subset $L \subseteq \Sigma^*$.

The empty language (set) is denoted $\emptyset$.

Examples of Languages

$$L = a^* =$$

$$L = 01^* =$$

$$L = (01)^* =$$

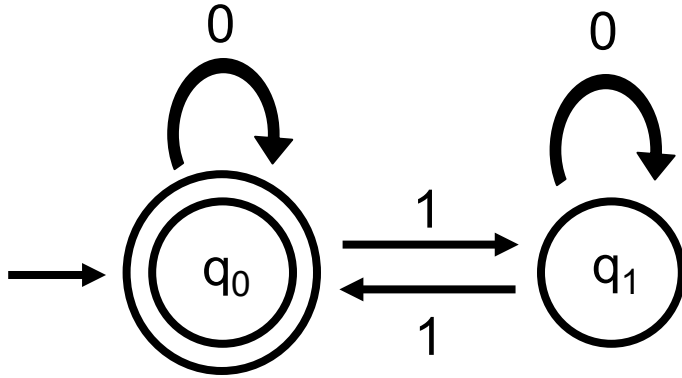
$$L = \{x^k y^k : k \geq 0\} =$$

Question: Suppose $L = \{\epsilon\}$. Does $L = \emptyset$?

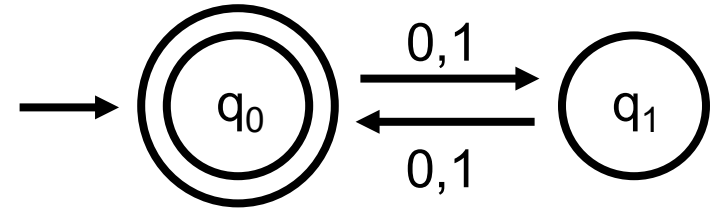
Question: Does a^* have infinite strings? infinitely many strings?

Question: (How) is $a^*|b^*$ different from $(a|b)^*$?

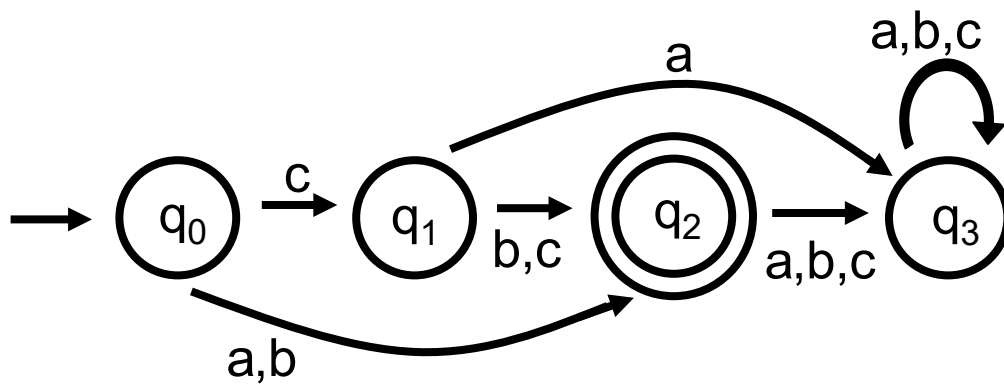
What languages do these 4 DFAs decide?



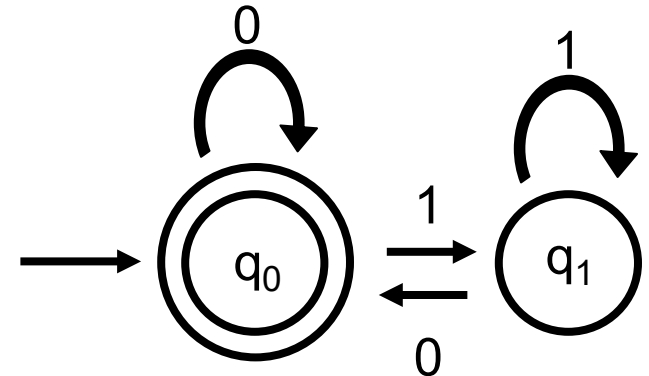
$L(M_1) =$



$L(M_2) =$



$L(M_3) =$



$L(M_4) =$

Formal definition of DFAs

A **deterministic finite automaton** is a 5-tuple:

$$M = (\Sigma, Q, q_0, F, \delta)$$

1

2

3

4

5

Formal definition of DFAs

A **deterministic finite automaton**
is a 5-tuple:

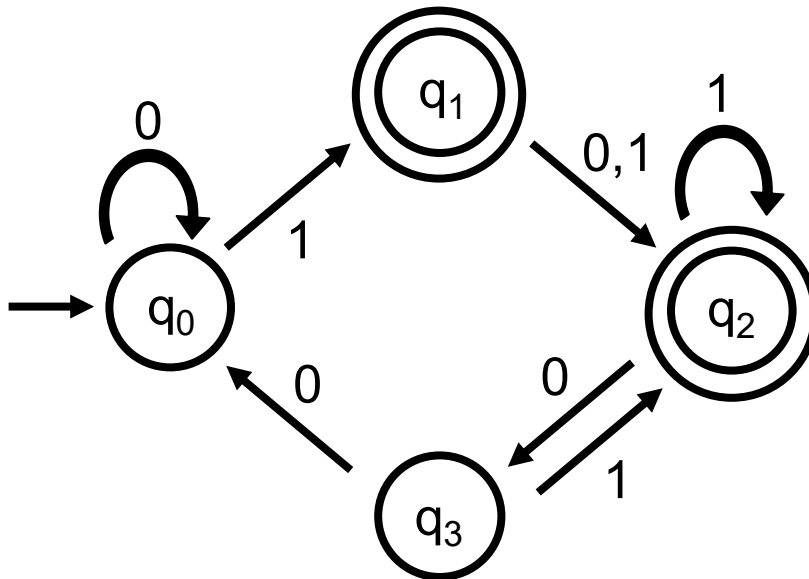
$$M = (\Sigma, Q, q_0, F, \delta)$$

$\Sigma =$

$Q =$

q_0

$F =$



δ	0	1
q_0		
q_1		
q_2		
q_3		

1. For $\Sigma = \{a,b\}$, consider the language of strings with exactly one b
 - a. Write using string notation: _____
 - b. Draw a DFA

2. Draw a DFA for the language of strings that have “aaaa” in them

3. Draw a DFA for $L = \{x^k y^k : k \geq 0\}$

Impossibility Proof

Consider *any* DFA **M**. We show **M** does not decide $L = \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}} \underline{\hspace{1cm}}$

Let $s =$ _____. (It's a finite number!)

Consider the (finite) input string $\mathbf{x^s y^s}$.

As we process the string, the DFA switches to some different states q_i :

ε	x	xx	xxx	$xxxx$	$xxxxx$	$\dots$	x^s
q_{i_0}	q_{i_1}	q_{i_2}	q_{i_3}	q_{i_4}	q_{i_5}	$\dots$	q_{i_s}



There are _____ (pigeonholes) and _____ (pigeons),
so there must be _____ such that _____.

Claim. The two inputs _____ and _____ have the same final state. Why?

This means **M** must

But $\in L$ and $\notin L$, so **M** is wrong on one of these!

Conclusion: M does not decide L .